Lab Experiment – 6

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Aim:

To design and verify the operation of a 4-bit Common Bus System using multiplexers/basic logic gates for transferring binary data between multiple registers.

Theory:

A Common Bus System is a digital communication pathway that allows data transfer among various registers or components through a shared set of lines. In a 4-bit Common Bus System, four parallel lines are used to transfer 4-bit data from one selected register to another.

The bus reduces the need for multiple direct connections between registers, simplifying circuit design and improving hardware efficiency.

Components Used:

- Multiple 4-bit registers (e.g., Register A, B, C, D)
- Multiplexers or tri-state buffers
- Selection lines
- Control signals

Working Principle:

- Each register contains 4 bits of binary data
- Selection lines determine which register's data is placed on the common bus
- Only one register can transfer data onto the bus at a time
- The selected data is then available for transfer to another destination register

Example:

If Register B is selected:

- Bus Output = B3 B2 B1 B0

If Register D is selected:

- Bus Output = D3 D2 D1 D0

Bus Structure:

For each bus line:

One multiplexer selects one bit from corresponding bits of all registers

Thus:

- Bus line 0 selects from A0, B0, C0, D0
- Bus line 1 selects from A1, B1, C1, D1
- Bus line 2 selects from A2, B2, C2, D2
- Bus line 3 selects from A3, B3, C3, D3
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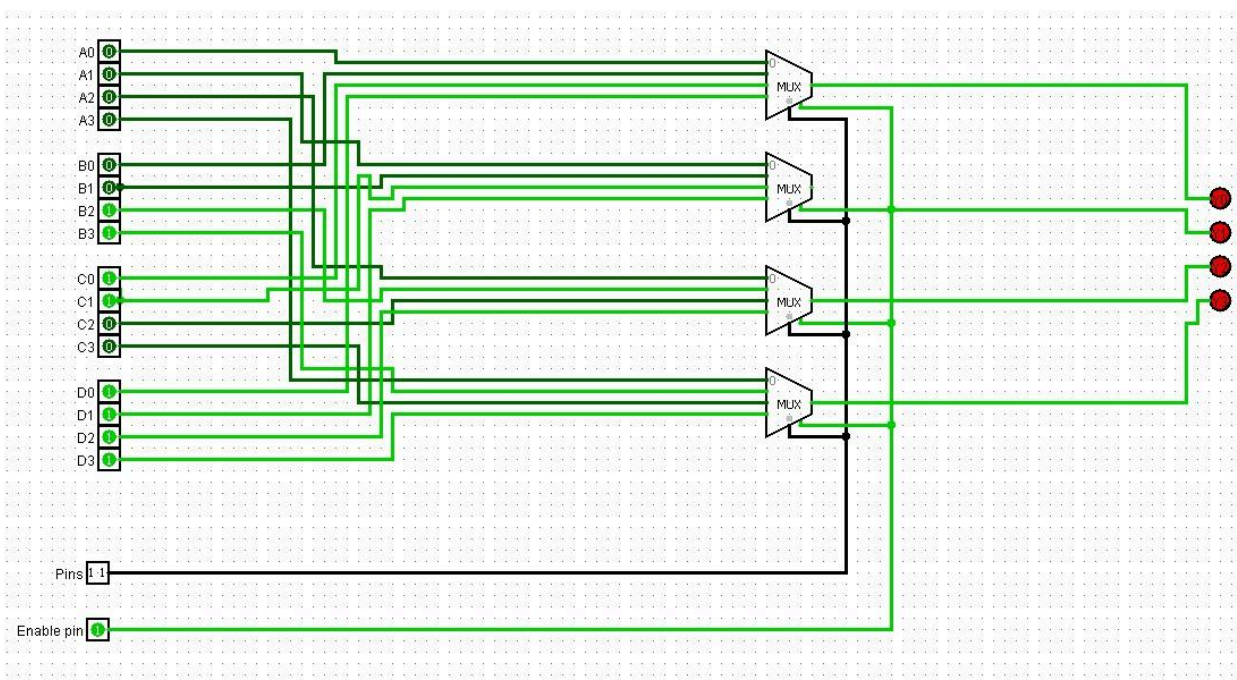
Logisim Implementation:

The 4-bit Common Bus System was designed using multiplexers and logic gates in Logisim, where different registers were selected using control lines to observe correct data transfer on the shared bus.



4- bit Common Bus Architecture